

Evaluating NYC DOT Response Efficiency Using 311 Complaint Data

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Overview

This project analyzes New York City 311 service request data, focusing specifically on Department of Transportation (DOT) complaints, to evaluate how efficiently the city responds to infrastructure-related issues. The goal of this analysis is to identify patterns in complaint volume and resolution time, uncover any inefficiencies, and provide some data-driven recommendations to improve service across boroughs.

This dataset was sourced from NYC Open Data and filtered to include valid, closed DOT complaints from 2024 to 2025. Using Excel and Google Sheets for data cleaning and Tableau for visualization, we explored trends in complaint types, geographic distribution, and response efficiency.

Key Findings

Our analysis revealed several important insights regarding the NYC DOT's performance:

- **Resolution time is primarily driven by complaint type.** Infrastructure-related issues such as Wayfinding, Bike Rack Condition, and Missing Street Signs take significantly longer to resolve (30+ days on average), while simpler issues like Public Toilets are resolved much faster.
- **Complaint volume is concentrated in a few categories.** Street Condition, Traffic Signal Condition, and Street Light Condition make up the majority of all complaints, indicating specific, recurring infrastructure challenges across the city.
- **Borough-level efficiency differences are relatively small but meaningful.** Average resolution times across boroughs are consistent, but Queens consistently shows slower response times compared to Manhattan and other boroughs.
- **High complaint volume doesn't necessarily lead to slower resolution times.** While Queens and Brooklyn handle the highest number of complaints, the overall resolution times remain consistent, suggesting that operational efficiency is influenced more by complaint complexity than volume.
- **Extreme outliers significantly impact average resolution times.** Some complaint types show large gaps between the average and median resolution times, indicating that a small number of cases experience unusually long delays, which skew the overall efficiency metrics.

Recommendations

Based on these findings, we propose the following recommendations:

1. **Allocate resources strategically to high-demand areas.** Increase staffing in boroughs like Queens and Brooklyn, where complaint volume and delays are highest.
2. **Prioritize high-impact infrastructure issues.** Focus process improvements on complaint types with consistently long resolution times to reduce delays and improve consistency.
3. **Investigate and reduce extreme delays.** Identify and analyze outlier cases, such as Street Condition in Staten Island to uncover any bottlenecks and improve consistency in resolution times.
4. **Use historical data to better plan and distribute workload across boroughs.** Analyze past trends to anticipate demand and distribute workload more effectively across boroughs, improving response times.

Conclusion

Overall, the NYC DOT demonstrates relatively consistent response times across all boroughs, but there are key inefficiencies remaining at the complaint-type level. Long resolution times are driven more by complexity and occasional extreme delays than by the overall workload. By focusing more on high-impact complaint categories, improving resource allocation, and addressing any outlier cases, the city can significantly enhance service efficiency and improve the experience of NYC residents.